

University of South Florida CDA 3201 Logic Design Lab Report

Semester: Fall 2022		
Experiment	Number	4
	Date	02/08/2023
Lab	Section	
	Instructor	Gustavo Olenscki
Report	Due Date	02/17/2023
Team	Member #1 name	Austin Molketin
	Member #2 name	Diya Jain

Purpose and objectives:

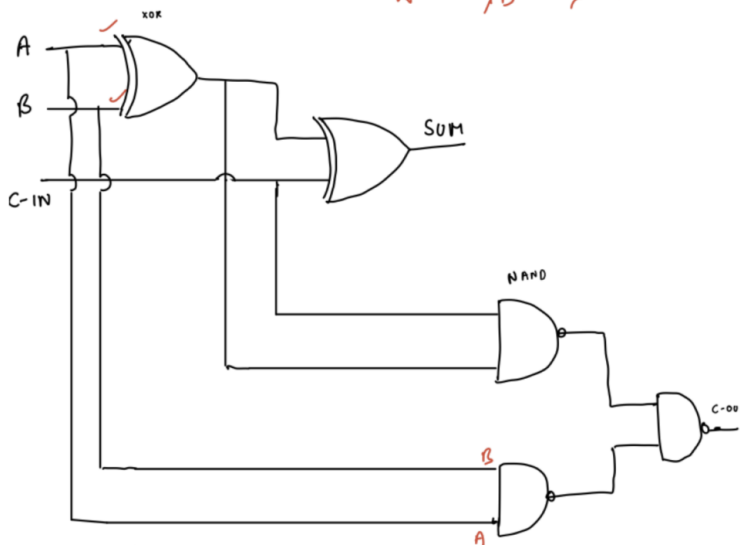
The purpose of this lab was to introduce us to basic modeling. combinational circuits using a hardware description language (HDL). Also, to use the knowledge of Verilog to verify our findings.

Components used:

We used the following components to complete the lab assignment:

- Breadboard
- 74SL00 Quad NAND gate
- 4LS86 Quad XOR gate
- Many wires
- The power unit
- 2 LED bulbs

Design description:



The design of the circuit was comparatively simple, we used NAND gates and XOR gates to complete this circuit. We connected the Vcc and GND to positive and negative respectively and created the whole circuit. Refer to the Image XX to see the circuit.

Observations and data analysis:

For this lab experiment we were provided with several questions.

Question 1: Derive the truth table and obtain the MSOP expression for sum and c-out using a k-map for each output variable. You should have two expressions, one for sum and one for c-out.

Answer:

A	B	C-IN	SUM	C-OUT
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

→ MSOP for sum and Cout

K-map for SUM

	BC			
	00	01	11	10
A' 0	0	1	3	2
A 1	1	5	7	6

$$A'B'C' + A'B'C + ABC + A'BC'$$

$$= A'B'C + A'BC' + AB'C' + ABC$$

$$= A \oplus B \oplus C$$

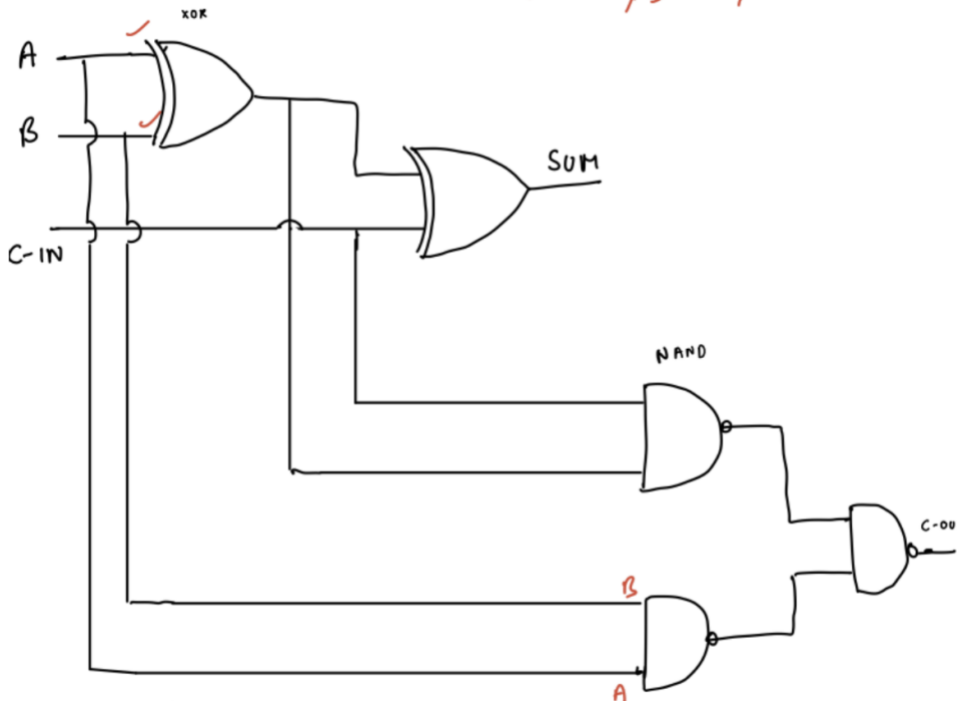
K-map for Cout

			1	
	0	1	1	0
1	1	1	1	1

$$= AC + BC + AB$$

Question 2: Show how sum can be implemented using XOR gates, and c-out can be implemented as a nand-nand network.

Answer:



Question3: This design can be optimized using Boolean algebra, but the derivation is somewhat complex (it requires adding in terms, distributing, and factoring multiple times). Instead, examine your truth table for c-out. What are the values of a, b, and c-in when c-out = 1?

Answer:

for $C = \sum m(3, 5, 6, 7)$ the value of Cout was 1.

Question 4: Implement the same optimized circuit using Verilog HDL, as discussed in the prelab. Implement a testbench which exercises all inputs. Confirm functionality by running the simulation and observing the output in the console as well as the waveform.

Answer:

```
2 // Or browse examples
3 `timescale 1ns / 1ps
4
5 module fullAdder_tb;
6
7
8 // Inputs
9
10 reg In1;
11
```

<https://edaplayground.com/x/N9jk>

10/7/22, 8:26 PM

(1) - EDA1

```
12 reg In2;
13
14 reg Cin;
15
16
17 // Outputs
18
19 wire Sum;
20
21 wire Cout;
22
23
24 //Temporary looping variable
25
26 reg [2:0] i = 3'd0;
27
28
29
30
31 // Instantiate the Unit Under Test (UUT)
32
33 fullAdder uut (
34
35     .In1(In1),
36     .In2(In2),
37     .Cin(Cin),
38     .Sum(Sum),
39     .Cout(Cout)
40 );
41
42
43
44
45 );
46
47 initial begin
48
49     // Initialize Inputs
50     $dumpfile("dump.vcd");
51     $dumpvars;
52     $display("Starting simulation...\n");
53     $display("ta\ttb\ttc\n");
54     $monitor("t%d\t%d\t%d", In1, In2, Cin);
55
56     In1 = 1'b0;
57     In2 = 1'b0;
58
59
```

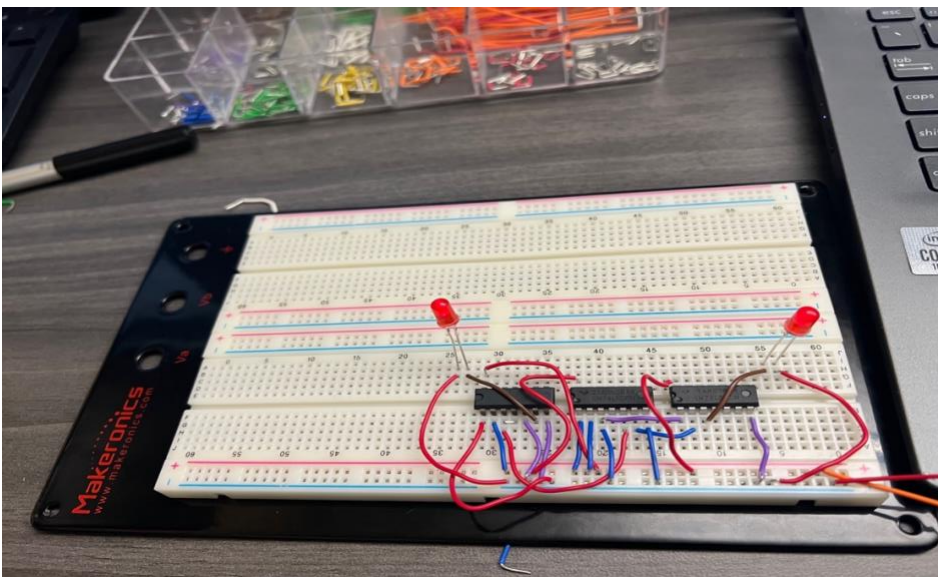
```
58 In2 = 1'b0;
59
60 Cin = 1'b0;
61
62
63 // Wait 100 ns for global reset to finish
64 #100;
65
66
67
68
69 // Add stimulus here
70
71 for( i= 0; i < 8; i = i + 1'b1)begin
72     {In1,In2,Cin} = {In1,In2,Cin} + 1'b1;
73
74     #20;
75
76 end
77
78
79 end
80 endmodule
```

<https://edaplayground.com/x/NQit>

```

1 // Code your design here
2 `timescale 1ns / 1ps
3
4 module fullAdder ( In1,
5
6     In2,
7
8     Cin,
9
10    Sum,
11
12    Cout
13 );
14
15
16 input  In1,
17
18    In2,
19
20    Cin;
21
22 output Sum,
23
24    Cout;
25
26
27 assign Sum = (In1 ^ In2) ^ Cin;
28
29 assign Cout = (In1 & In2) | (In2 & Cin) | (Cin & In1);
30
31 endmodule

```



Discussion and conclusion:

In conclusion, the objective of this lab was to introduce us to modeling basic combinational circuits using a hardware description language (HDL) and also to use the knowledge of Verilog to verify our findings. we were able to complete all parts of the lab

experiment successfully. Throughout the experiment we followed all the safety procedures advised by the TA.

Answers to questions:

Some extra questions that are needed to be answered:

Question 1: In Step 5, you demonstrated the circuit functionality by testing different combinations of inputs. What are the potential challenges of testing larger circuits with more inputs using this technique?

Answer: Testing a circuit with more inputs can be difficult because more combinations of inputs need to be tested. Additionally, some inputs may not be connected to anything, which can make testing more difficult.

The potential challenges of testing larger circuits with more inputs using this technique are that it can be difficult to keep track of all the potential combinations of inputs, and it can be difficult to test all the possible inputs.